

NOISEMAP · REQUIREMENTS

NoiseMap requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

NoiseMap requirements

These requirements are proposed targets for concept review, awaiting Amish, and must be revised with a co-design partner before the design is frozen. Status is judged against the first-order estimates in NSM-PRC-001; "met" means met on paper only. Two requirements are **not met** by the current concept (R11 wind flagging and R13 cost), one is met only in part (R12 at slow data rates), and three are at risk (R3, R4 at the low end, R5 at low frequencies).

Design case: one node on a street pole 0.45 m behind the curb, microphone about 4 m above the ground and 0.45 m from the pole face, in a mixed street with traffic by day and bars at night.

Table 1. Requirements.

ID	Requirement	Target	Verification	Concept status
R1	Privacy: no audio samples are stored or leave the microphone head; only levels cross the cable or the radio	No data storage in the head; cable carries levels only; head firmware published with a build hash	Design review, firmware audit, cable capture	Met by design; see NSM-PRC-001 on the limits of the physical argument
R2	Reported metrics	Per 15 min: 1-minute LAeq, LAeq, LAFmax, LAFmin, L10, L90, LCeq, device status	Design review	Met by design (22 bytes per record)
R3	Accuracy of LAeq,15min against a co-located class 1 meter	Within ± 2 dB from 40 to 100 dBA, after field calibration	Side-by-side field test	At risk: housing, windscreen and reflections unverified
R4	Measuring range	35 to 110 dBA	Bench test with calibrator and reference source	Upper end met (about 115 dBA); at risk at 35 dBA (self-noise about 29 dBA)
R5	Frequency weighting	A and C weighting as defined in IEC 61672-1, within class 2 tolerances from 63 Hz to 8 kHz (goal)	Bench test against a reference meter	At risk below about 60 Hz (microphone roll-off)
R6	Time stamps	Each record time-stamped within 2 s of UTC	Design review (LoRaWAN network time)	Plausible
R7	Energy	Energy neutral at 1.5 peak sun hours; 14 days without sun	Energy budget, bench test	Met (about 5.8 Wh/day for 0.51 Wh/day; about 30 days)
R8	Ingress and weather	IP65 enclosure; microphone head survives driving rain; -20 °C to +50 °C	Spray test, thermal chamber	By design, unverified
R9	Calibration check	Field check with a 94 dB, 1 kHz calibrator in 10 min or less; offset stored on the node	Timed trial	Plausible
R10	Installation and mass	Two trained people, 45 min, no drilling or pole wiring; 3 kg or less; withstands 35 m/s gusts	Timed trial, weighing, bracket calculation	Met on mass (about 2.3 kg); wind load about 50 N to check
R11	Contaminated data	Intervals with wind above about 5 m/s or heavy rain are flagged	Field comparison with a weather station	Not met: no wind or rain sensing in the concept
R12	Radio use	Within EU868 1 % duty cycle and The Things Network fair use (30 s uplink per day)	Airtime calculation	Met at SF9 or faster (about 24 s/day); not met at SF10 without 30-minute records
R13	Parts cost	\$100 or less per node	Priced BOM	Not met: about \$171 (NoiseMap-specific parts about \$52)

ID	Requirement	Target	Verification	Concept status
R14	Service life	5 years outdoors, with a yearly windscreen change and one battery change	Design review, UV-stable materials	Unverified
R15	Openness	Hardware, firmware, calibration method and data format published under the repo licenses	Design review	Met by design

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Microphone height	about 4 m above the ground	4.0 ± 0.2 m assessment height for EU strategic noise maps (Directive 2002/49/EC, Annex I)
Quietest night level to resolve	about 35 dBA	Quiet residential street at night (estimate)
Loudest level to resolve	about 110 dBA	Close pass of a loud motorcycle or siren (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Wind speed that spoils readings with a foam windscreen	about 5 m/s	Common practice (estimate, to be checked)
Microphone data	65 dB SNR, 120 dB SPL overload	ICS-43434 product page (TDK InvenSense)

Safety: Requirements R8 and R10 involve work at height beside traffic and a lithium cell. See the safety section of NSM-PRC-001.